

A Novel Stack Ensemble Learning Techniques for Sentiment Classification in Climate Change Discourse

Vipin Jain^{a, *}, Shilpa Agnihotri Pandey^{b, **},
Ankit Chakrawarti^{c, ***}, Rahul Sharma^{d, ****}, and Vinod Patidar^{e, *****}

^a SCSE, VIT Bhopal University, Madhya Pradesh, 466114 India

^b Department information technology, Pimpri Chinchwad College of Engineering and Research,
Ravet, Pune, 411057 India

^c Department of Computer Science & Information Technology Chameli Devi Group of Institutions, Indore, India

^d Department of Artificial Intelligence and Machine Learning Manipal University Jaipur, Jaipur, India

^e Department of Computer Science and Engineering Parul Institute of Technology, Parul University,
Vadodara, Gujarat, India

*e-mail: er.vipinjain@gmail.com

**e-mail: shilpa.agnihotri28@gmail.com

***e-mail: chak03ankit@gmail.com

****e-mail: sharma.rahul5656@gmail.com

*****e-mail: vinod.patidar28579@paruluniversity.ac.in

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Abstract—Primary reason of climate reason is release of greenhouse gases into the biosphere which warms the earth's atmosphere. Climate change is serious issue that affects public sentiment significantly. In this paper, stack ensemble learning based framework is proposed to analyze public sentiment regarding climate change by using text datasets. First step of proposed framework is text preprocessing which includes tokenization, stemming, and stop-word removal to create TF-IDF representations. Next, stack ensemble model is developed for sentiment classification as positive, neutral, or negative. This model utilizes four machine such as support vector machine (SVM), decision tree (DT), random forests (RF), and gradient boosting classifier (GDB) as base models. The predictions given by all four models are combined using XGBoost as a meta-classifier. The four dataset related to climate change are used for experimental analysis. Results show that the proposed model achieves 89.72% precision, 91.65% recall, 90.23% F1-score, and 91.47% accuracy. These results suggest that the stack ensemble learning approach is effective for analyzing public sentiment about climate change.

Keywords: stack ensemble learning, sentiment analysis, climate change, XGBoost, NLP, machine learning

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1. INTRODUCTION

Global warming is one of the most serious environmental problem of the 21st century [1]. This is caused by greenhouse gases in the atmosphere, which trap heat from the sun [2]. Human activities like burning fossil fuels release greenhouse gases like methane and carbon dioxide into the atmosphere. In addition to trapping heat in the atmosphere, these gases are contributing to an increase in global temperatures. Increasing temperatures can cause changes natural disasters, extreme weather events, and weather patterns. These impacts seriously affect ecosystem, human health, and global economy. It is essential to address climate change now in order to protect our planet. In recent years, machine learning techniques are applied to analyze public sentiments about climate change.

Millions of people share their opinions daily on Twitter, blogs, and wikis. Various social media messages posted by provide insight into their attitudes and feelings [3]. Social media data can be used for sentiment analysis. It is an integral part of natural language processing (NLP) [4]. Various techniques and methodologies are used in NLP to enable computers to understand, interpret, and generate human language. Sentiment analysis aims to obtain insights from unstructured text using advanced algorithms and linguistic models. These insights help businesses, researchers, and individuals to understand public

sentiment, customer satisfaction, and brand perception. following techniques can be used for sentiment analysis [5].

- **Lexicon-based:** in this technique list of words are associated with positive, negative, or neutral sentiment. This list of words is used to analyze text and determine text sentiment. It is commonly used in NLP applications to understand text sentiment. This method is relatively accurate and efficient compared to other more complex methods. It requires manual curation, which is time-consuming and prone to human error. Additionally, lexicon-based sentiment analysis only works in a limited scope and is not able to capture the context of the sentence, which can be important in determining the sentiment. This can lead to inaccurate results.

- **Machine learning:** these algorithms are trained with set of text annotations already categorized as positive, negative, or neutral. After that, the algorithms learn the patterns associated with each sentiment in the text. It can be used to predict the sentiment of new text which allows businesses to monitor customer sentiment in real-time. This can be used to identify customer needs and respond quickly about their issues. This data can be used to improve customer service and product development. It can also help businesses to understand customer sentiment and make more informed decisions.

- **Deep learning:** deep learning techniques have recently become popular for sentiment analysis. Deep learning algorithms can be trained to recognize patterns in large datasets and generate insights regarding customer sentiment. Additionally, deep learning models detect subtle nuances in text which is not detected by conventional machine learning models.

This study examines the effectiveness of stack ensemble [6] learning model for sentiment analysis of tweets related to climate change. The purpose of this study is to ensemble multiple base models in order to achieve high sentiment classification accuracy. Ensemble learning is a machine learning paradigm that aggregates the predictions of multiple individual models to produce a more robust and accurate prediction.

1.1. Novelty of This Work

This work proposed an efficient sentiment classification model for climate change-related tweets using stack ensemble machine learning model. The primary significance of this paper is presented as:

- Implemented a stack ensemble learning approach for analyzing public sentiment about climate change.
- This approach combines the strengths of four base models: SVM, RF, DT, and GDB.

The structure of the remaining paper is as follows, Section 2 covers existing literature. Section 3 covers the proposed methodology used for sentiment categorization. Section 4 consists of implementation details with result analysis. Section 5 concludes with recommendations for future research.

2. RELATED WORKS

Sham et al. investigated the effectiveness of many sentiment analysis approaches on climate change tweets [1]. They used two feature extraction techniques to compare seven lexicon-based approaches and three machine learning classifiers. Finding that the hybrid approach of TextBlob and Logistic Regression outperformed the other two approaches, with highest F1-score of 75.3%. This study showed that TF-IDF outperformed BoW by 0.6%, whereas lemmatization improved machine learning and hybrid models by 1.6%. Merchan et al. (2023) fine-tuned BERT and ClimateBert, two transformer models, on the novel Clima-Text database using transfer learning. [7]. Their results showed that ClimateBert outperforms the other models and it can be used to monitor climate risk disclosure in financial reports. It is inexpensive regarding computational resources, so any organization can use this finding. Using Reddit posts and comments, Ray et al. analyzed users feeling about climate change [8]. The study found that Reddit users have a negative sentiment towards climate change, especially when discussing extreme weather events. CNN and RNN neural networks have been used to predict the sentiment of the posts. A CNN model performed slightly better than an RNN model regarding precision, accuracy, and loss. It has been suggested that social media can be used to raise climate change awareness. Cody et al. analyzed tweets containing the word “climate” to determine people’s views on global warming [9]. Studies have shown that people are unhappy when hearing about natural disasters, climate change legislation, or oil drilling. However, people tend to be happier when they hear about climate rallies or green ideas contests.

Upadhyaya et al. proposed a model named STASY to classify tweets related to climate change [10]. STASY incorporates temporal orientation and sentiment analysis as additional tasks to better understand the context and sentiment of the tweets. This framework classified tweets as either deniers or believers in their stance on climate change. Experiments show significant improvements in stance detection perfor-

mance across various datasets, including publicly available climate change datasets. The average improvement in F1 scores ranges from 12.14 to 35.01% compared to baseline methods.

Mi et al. conducted sentiment and emotion analysis using R tools on a corpus of tweets from 2015 to 2018 mi2023text. Their study aims to understand public attitudes and emotions on Twitter related to climate change topics. The study provides insights into Twitter users' opinions and emotions regarding climate change. Additionally, they emphasize the need for better sentiment analysis tools to understand public sentiment.

Using machine learning methods, Kirelli et al. analyzed Turkish tweets about climate change and global warming [11]. Thirty thousand randomly selected Turkish tweets have been trained using supervised learning algorithms (linear and probabilistic classifiers) to detect positivity, negativity, and neutrality in sentiments. Based on their findings, integrated classification methods perform more accurately than single machine learning techniques.

Wang et al. used the Twitter API and the Tweepy library to analyze the sentiment of tweets with #climate change hashtags [12]. Using multinomial naive Bayes classifiers, they trained a model to detect people's views about climate change with accuracy of 67%. The findings show that most people in the training group expressed a negative sentiment, and few are optimistic about climate change [13].

Barachi et al. proposed an approach to analyze social media posts in realtime [14]. Using this model, tweets have been classified into different categories, such as support, opposition, anger, inspiration, and joy. It is trained on a total of 278 000 tweets related to climate change and has achieved high accuracy in all categories.

- **Limitations and challenges of existing techniques:** Sentiment analysis is an area of NLP that automatically identifies the sentiments conveyed with words. Significant works is done by various authors in this field. Limitations and challenges of existing works are given as:

- **Use of single sentiment lexicon:** measures the sentiment of tweets about climate change, most studies use single sentiment lexicon such as SentiStrength lexicon. This is an issue because different lexicons can have different perspectives so the sentiment of tweets about climate change may not be captured correctly.

- **Use of single time period:** tweets related to climate change are analyzed in many studies over single time period, such as preceding or following a major climate event. However, a single time period may not capture the full scope of the public's reaction to climate change. Long-term studies are needed better to understand the public's engagement with climate change.

3. PROPOSED MODEL

The sentiment analysis framework proposed in this research has been structured into three fundamental phases: (1) data collection, (2) data preprocessing, and (3) sentiment analysis. A visual representation of the proposed methodology is presented in Fig. 1. The following subsequent sections provide details of each stage.

3.1. Data Collection

Four Twitter datasets related to climate change are used in this work to analyze the sentiments of people towards it. All related data is scraped from Twitter and annotated with sentiment polarity. All four datasets contain different numbers of tweets which are categorized as negative, neutral, or positive.

- **Weather sentiment dataset (Dsa-1):** It is first dataset used in this study downloaded from Data World website [15]. It contains 1000 weather related tweets. All tweets are categorized into four categories according to user's sentiments: negative, "I don't know," positive, and neutral/author is just sharing information. This study analyzed tweets containing neutral, negative, and positive sentiment polarities. After removing tweets with other sentiment polarities, 763 remaining tweets are used for further analysis.

- **Earth hour 2015 Corpus (Dsa-2):** It is second dataset contained 371 tweets obtained from the DecarboNet database [16, 17]. Tweets are annotated with sentiment polarity, but their contents are not provided. The Tweepy library is used to extract the contents of 371 tweets from the corpus, but only 252 tweets have been returned.

- **Kaggle dataset (Dsa-3):** It contains 396 tweets about climate change sentiments collected between January 1, 2020, and December 24, 2020 [18].

- **Tweeter dataset (Dsa-4):** Forth dataset of tweets related to climate change are collected using the Tweepy API from January 1 to March 31, 2023 from Tweeter. It contains total 31,961 tweets.

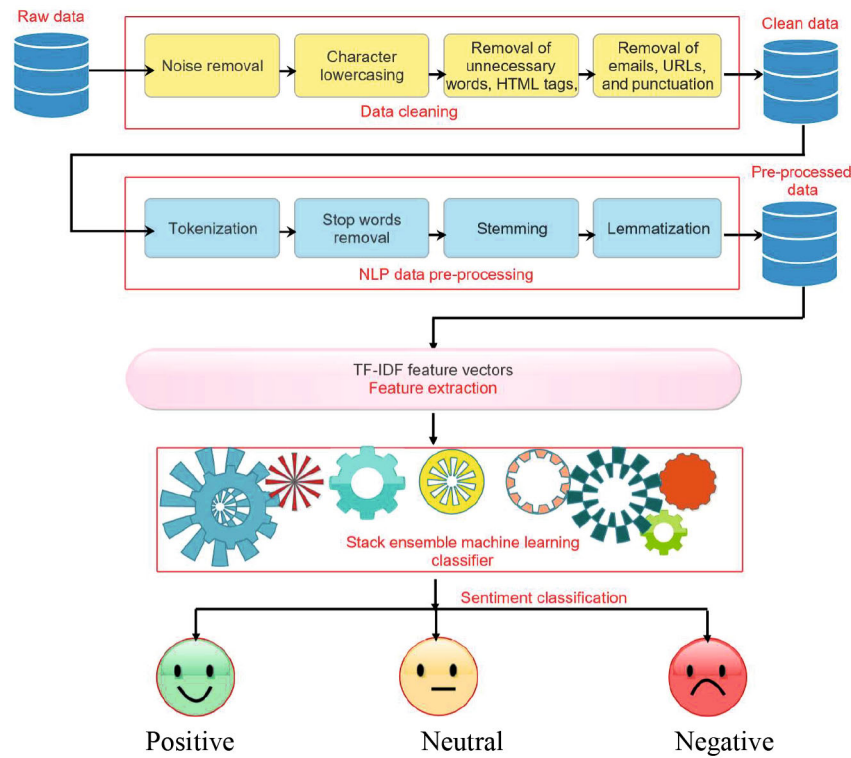


Fig. 1. Schematic diagram of the proposed framework.

3.2. Data Preprocessing

It contains cleaning and normalization step of data before sentiment analysis. This step helps to improve accuracy of sentiment analysis findings. Different tasks performed in this step are given as:

- **Data cleaning:** process removes various noises present in collected tweets including unnecessary words, emails, punctuation, URLs, blank spaces, HTML tags, and emoji [19].

- **Tokenization:** The tokenization process divides text strings into tokens. Each token consists of a sequence of characters with specific meaning. For example, the sentence “climate change is a real problem today we are globally facing” would be tokenized as follows: (“climate,” “change,” “is,” “real,” “problem,” “today,” “we,” “are,” “globally,” “facing”). Extracted tokens are stored as features for labeling to assign a sentiment label to each tweet. Sentiment labels can be either negative, positive, or neutral.

- **Stop words removal:** Stop words are common words which does not convey any sentiment. These word needs to be removed from the text before analyzed by NLP algorithm. There are different ways to remove stop words from text. One common method is to use a list of stop words to remove these words from text automatically. Removing stop words from text can improve accuracy of sentiment analysis algorithm.

- **Normalization of data:** Lemmatization and stemming strategies are employed for data normalization. Stemming process reduces a word to its root form. In this process, suffixes and prefixes of the word are removed. For example, given word *Changing* can be stemmed to *Chang* by removing the suffix *-ing*. The lemmatization process reduces word dimensionality by grouping similar words. For example, the words *running*, *ran*, and *runs* would all be lemmatized as *run*.

- **Data labelling:** In this step, unlabeled tweets are labelled by using VADER lexicon. It is a sentiment analysis lexicon commonly used in social media research. It is a rule-based lexicon that uses variety of linguistic features to determine the sentiment of a text, including word lists, negation, punctuation, and capitalization. The VADER lexicon calculates compound score for each text, which is a single number that reflects the overall sentiment of given text. Range of computed compound score is from -1 (very negative) to 1 (positive).

- **Vectorization:** Text data have been vectorized by using TF-IDF technique which allows for further classification. The TF-IDF algorithm extracts features from text based on word frequency. Words that

occur more frequently in one document and less frequently in another are given more weight. TF-IDF weight is calculated as follows [20]:

$$\text{TF-IDF} = \text{TF} * \text{IDF}. \quad (1)$$

TF is the number of times a phrase appears in a document; IDF is the inverse document frequency calculated by dividing the number of documents in the corpus by the number of sentences containing the phrase.

3.3. Classification Model

Sentiment classification of tweets about climate change has been performed by using six machine learning models: SVM, RF, DT, GBD, XGBoost, and proposed stack ensemble model. The following subsections provide brief descriptions of each model.

3.4. Support Vector Machine

This algorithm is suitable for both classification and regression [21]. It determines the best hyperplane to separate classes by mapping data into a high-dimensional feature space. This algorithm optimizes the margin and minimises training error. SVM determines a bias f and set of weights a for each hyperplane. The SVM algorithm can be expressed as an optimization problem and it is represented as:

$$\min_{a,f} \frac{1}{2} a^2 + g \sum_{m=1}^q \max(0, 1 - e_m(a^S p_m + f)), \quad (2)$$

where a is the weight vector; f indicates the bias; p_m denotes the m th data point; e_m is the label for the m th data point; g controls the trade-off between margin and number of misclassified data points.

3.5. Random Forest

It is powerful ensemble learners that integrates several decision trees to produce more accurate and generalized predictions. This model can handle large datasets. Additionally, RF model is relatively easy to understand and interpret. This model trains decision trees with randomly selected subsets of features. It makes predictions by calculating average of all decision trees. Compared to a single decision tree, this model reduces prediction variance and bias [22]. Furthermore, this model is less susceptible to over-fitting and can handle more complex datasets. Mathematical description of RF algorithm is given as: several

$$\text{RF}(x) = \operatorname{argmax}_{c \in C} \sum_{i=1}^n I(T_i(x) = c), \quad (3)$$

where x , C denotes tweet and set of possible sentiment labels, respectively; $T_i(x)$ denotes prediction of i th decision tree for x ; $I(T_i(x) = c)$ denotes an indicator function, that is, 1 if $T_i(x) = c$ and 0 otherwise.

3.6. Decision Tree

It is one of the base model of proposed stack ensemble learning approach [23]. It is powerful nonlinear model widely used for classification tasks. DT is well-suited for determining whether sentiment is positive, negative, or neutral. Decision tree is represented mathematically as a hierarchical structure of nodes that recursively split the data into subsets based on different features. It is mathematically represented as:

$$\text{DT}(x) = \max_{c \in C} P(c) \prod_{i=1}^n P(x_i | c), \quad (4)$$

where x denotes tweet; C denotes set of possible sentiment labels; $P(c)$ represents prior probability of class c ; $P(x_i | c)$ is the probability of feature x_i in given class c [24].

3.7. Gradient Boosting Classifier

GDB is model which combines the predictions of multiple models to make a final prediction. In this model, decision trees are repeatedly added. The decision trees are trained to correct errors made by previous decision trees. This process is repeated until model achieves convergence. This model is denoted as

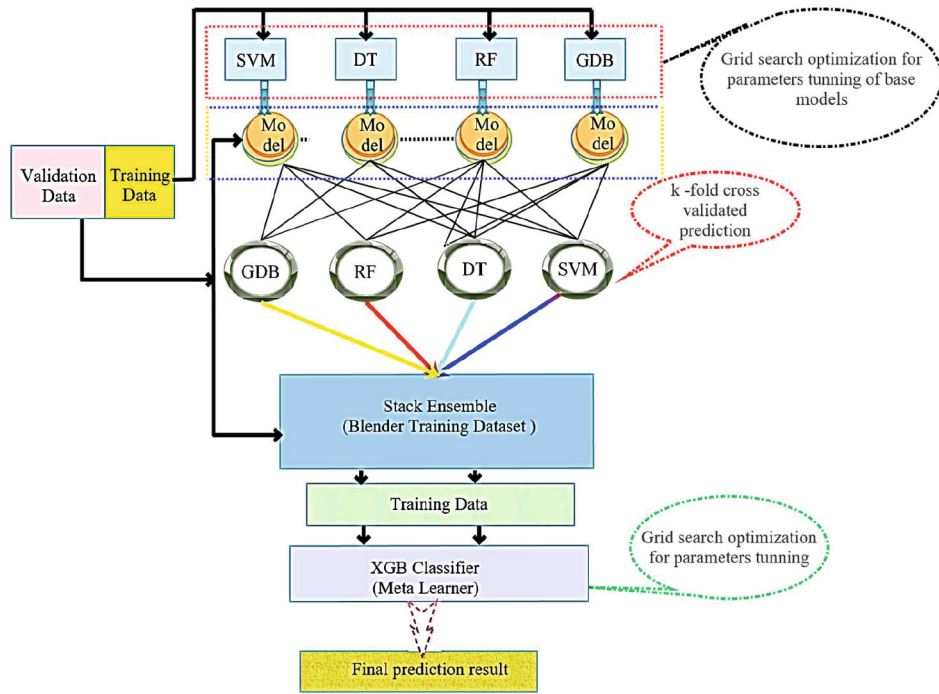


Fig. 2. Proposed stack ensemble machine learning architecture.

$$GB(x) = f(x) = H_0(x) + \sum_{i=1}^l \beta_i h_i(x), \quad (5)$$

where $GB(x)$ denotes output of GDB model; $x, f(x)$ denotes tweet, and predicted sentiment label of x ; $H_0(x)$ is base model; $h_i(x)$ denotes i th decision tree; β_i is the weight of the i th decision tree.

3.8. XGBoost Classifier

XGBoost is an optimized version of GDB algorithm used for classification tasks [25]. It is fast, accurate, and robust algorithm. The mathematical equation for the XGBoost algorithm is shown here:

$$GBDT(x) = f(x) = H_0(x) + \sum_{i=1}^T \beta_i h_i(x), \quad (6)$$

where $GBDT(x)$ is the XGBoost model x is a tweet, $f(x)$ is the predicted sentiment label for x ; $H_0(x)$ is the base model, $h_i(x)$ is the i th decision tree; β_i is the weight of the i th decision tree.

3.9. Proposed Stack Ensemble Model

The stack ensemble model has been developed for categorization of sentiment. The proposed ensemble model is illustrated in Fig. 2. It uses four base models: SVM, RF, DT, and GDB. Prediction of all four base models are combined using XGB classifier as a metaclassifier. All base models are trained with same training data. However, each base model has a different set of parameters. All parameters of the base models are tuned using grid search technique. The best set of parameters for each base model is used to train the ensemble model. The XGB classifier is trained by using predictions of the base models to make a final prediction.

4. EXPERIMENTAL SETUP

The results of each step of proposed models are discussed in this section.

4.1. Data Preprocessing Results

This involves converting raw texts into formats that machines can easily understand and process. It includes tokenization, removing stop words, and lowercase. After preprocessing input text, TF-IDF val-

Table 1. Sample of preprocessing steps and its output

Steps	Output
Original text	"Climate change is real problem today we are globally facing. "
Cleaned text	climate change is real problem today we are globally facing
Tokenization	['climate', 'change', 'is', 'real', 'problem', 'today', 'we', 'are', 'globally', 'facing']
Removing stop words	[climate change real problem today globally facing]
Lemmatization	climate change real problem today global face
Vectorization	[0.30504, 0.305304, ... , 0. , 0. , 0. , 0.181146, 0. , 0. , 0.30504, 0.305304],

Table 2. Sentiment distribution of Tweets in datasets

Dataset Name	Total tweets	Neutral	Positive	Negative
Dsa-1	763	261	231	271
Dsa-2	252	162	64	26
Dsa-3	1411	549	485	377
Dsa-4	31961	5022	10227	16712

ues can be calculated for each word in each document as shown in Table 1. The TF-IDF value represents the importance of word in a document compared to entire corpus.

4.2. Results of Sentiment Analysis

Table 2 shows tweet classification results based on their emotional assessment of several datasets, including weather, Earth Hour 2015, and climate change. In sentiment analysis, tweets are classified into three categories: neutral, positive, and negative. Following is an overview of the table information:

The sentiment analysis results for each dataset are as follows:

Dsa-1: Sentiment analysis is conducted on total 763 tweets of this dataset. Total 261 tweets are categorized as neutral, 231 tweets as positive, and 271 tweets as negative sentiment.

Dsa-2: This dataset contains 252 tweets. Out of these, 162 tweets are classified as neutral, 64 tweets are categorized as positive, and 26 tweets are categorized as negative.

Dsa-3: This dataset consists of 1411 tweets for sentiment analysis. Among these, 377485, and 549 tweets are classified as negative, positive, and neutral respectively.

Dsa-4: This dataset contains 31961 tweets collected from Twitter between January 1 and March 31, 2023. The tweets are collected using a various keywords and hashtags related to climate change. Among these, 16712, 10227, and 5022 tweets are labelled negative, positive, and neutral respectively.

4.3. Classification Results

The performance of the ensemble hybrid model and each individual classifier is evaluated using four metrics: recall, precision, F1-score, and accuracy.

• **Recall:** it is a metric of a classifier's correctness. It refers to the percentage of positive instances that have been classified correctly. It is calculated by using the following expression:

$$\text{Recall} = \text{TruePositives} / (\text{TruePositives} + \text{FalseNegatives}). \quad (7)$$

True positives are the cases that are accurately identified as positive. False negatives are cases that are incorrectly identified as negative. A high recall indicates that the classifier did not miss many positive instances.

• **Precision:** It is a measure of the accuracy of a classifier. This is the percentage of positive predictions that are actually true. It is calculated by using the following mathematical expression:

$$\text{Precision} = \text{TruePositives} / (\text{TruePositives} + \text{FalsePositives}). \quad (8)$$

False Positives are instances that have been incorrectly classified as positive. A high precision indicates that the classifier did not make many false positives.

Table 3. Performance evaluation (in %) of different classifiers on Earth Hour 2015 dataset

Models	Precision	Recall	F1-score	Accuracy
RF	80.23	82.56	81.87	83.74
DT	78.92	76.45	76.73	78.12
SVM	76.32	78.27	76.85	75.21
GDB classifier	81.46	83.57	85.22	86.38
XGB classifier	86.17	85.74	76.52	86.25
Stack classifier	85.45	86.25	84.57	86.89

Table 4. Performance evaluation (in %) of different classifiers on Weather Sentiment dataset

Models	Precision	Recall	F1-score	Accuracy
RF	77.89	76.59	78.11	80.47
DT	79.51	78.63	80.74	81.75
SVM	80.45	82.71	81.68	84.25
GDB classifier	84.65	82.17	83.97	83.56
XGB classifier	82.57	84.36	84.71	86.83
Stack classifier	89.32	97.45	88.65	89.77

• **F1-score:** It measures accuracy and completeness of classifier. Mathematically, it is calculated as follows:

$$\text{F1-score} = 2 * (\text{precision} * \text{recall}) / (\text{precision} + \text{recall}). \quad (9)$$

Here, precision measures how many positive predictions are actually correct. High F1-score indicates that the classifier is both accurate and precise.

• **Accuracy:** It is the overall performance of classifier's [26]. It is proportion of all instances that are correctly classified. It is calculated using the following mathematical expression:

$$\text{Accuracy} = (\text{TruePositives} + \text{TrueNegatives}) / \text{TotalInstances}. \quad (10)$$

True negatives are cases that are accurately identified as negative. A high accuracy indicates that the classifier correctly classified most of the instances.

Performance comparison of different classifiers on the Earth Hour 2015 dataset is shown in Table 3. The Stack classifier achieved highest accuracy of 86.89%. The GDB, XGB, and RF classifier obtained comparable accuracy scores of 86.38, 84.52, and 83.74%, respectively. The DT classifier achieved lowest accuracy of 78.12%. The lowest score is obtained for all four metrics by SVM classifier. The stack classifier achieved highest precision and F1-score for all three classes. Performance of various classifiers on Weather Sentiment dataset is summarized in Table 4. The stack classifier achieved highest accuracy of 89.77%, followed by the GDB classifier with 83.56% accuracy. XGB and SVM classifiers achieved 86.83 and 84.25% accuracy, respectively. In all four metrics, the RF achieved the lowest performance with lowest accuracy of 81.75%. Additionally, highest precision and F1-score is achieved for all three classes by stack classifier.

Table 5 shows that the Stack classifier achieved the highest accuracy of 86.68%, followed by the SVM classifier with 80.54% accuracy. The GDB and RF classifier gives comparable accuracy scores of 80.47 and 77.89%, respectively. Lowest value is obtained by DT and XGB classifier for all four metrics. Highest precision and F1-score is obtained by Stack classifier for all three classes. Table 6 presents the performance evaluation of different classifiers on tweets dataset. The stack classifier achieved the highest accuracy of 91.47%, followed by GDB with 86.91% accuracy. The XGB and the SVM classifier had comparable accuracy scores of 88.32 and 83.09%, respectively. Lowest performance is achieved by DT and RF classifier for all four metrics.

The proposed stack classifier performed best with all four datasets with highest accuracy of 86.89, 89.77, 86.68, and 91.47% Earth Hour 2015, Weather Sentiment, Climate Change Sentiment, and Twitter dataset, respectively.

Table 5. Performance evaluation (in %) of different classifiers on Climate Change Sentiment dataset

Models	Precision	Recall	F1-score	Accuracy
RF	78.19	76.55	79.12	77.89
DT	81.57	83.19	82.38	79.65
SVM	81.25	82.57	76.92	80.54
GDB classifier	77.59	76.11	80.36	80.47
XGB classifier	82.19	81.36	83.56	84.87
Stack classifier	88.65	88.92	87.45	86.68

Table 6. Performance evaluation (in %) of different classifiers on Tweeter dataset

Models	Precision	Recall	F1-score	Accuracy
RF	80.47	83.18	82.56	81.32
DT	82.36	83.54	85.71	83.58
SVM	86.39	85.64	82.4	83.09
GDB Classifier	88.74	86.24	85.37	86.91
XGB Classifier	85.63	86.25	87.46	88.32
Stack Classifier	89.72	91.65	90.23	91.47

Table 7. Comparison of suggested model to previous work

Reference	Approach/Model	Accuracy
Sham et al. [1]	Logistic regression and TextBlob with TF-IDF	75.3%
Kirelli et al. [11]	KNN	74.63%
Wang et al. [12]	Naive Bayes	67%
Barach et al. [14]	Bi-LSTM	89.80%
Proposed model	Stack ensemble	91.47%

4.4. Comparison with Existing Model

Comparison of proposed model with current work is depicted in Table 7. Sham et al. [1] applied combination of lexicon and machine learning-based methods to analyze climate change sentiment on Twitter. They have used seven lexicon-based approaches, three machine-learning classifiers, and two feature extraction techniques. Combining lexicon and machine learning techniques outperformed the other two approaches, achieving an F1-score of 75.3%. Kirelli et al. [11] used machine learning methods to analyze the sentiments of Turkish tweets regarding climate change and global warming. Authors achieved highest accuracy of 74.63% with K-NN algorithm. Wang et al. [12] conducted sentiment analysis of Twitter data to understand people's opinions on climate change over the years. They used four types of methodologies: rule-based, statistical, lexicon-based, and machine learning. Highest accuracy of 67% is achieved with machine learning algorithm.

Barach et al. [14] used Bi-LSTM classifier to analyze climate change sentiment on Twitter. The classifier is trained with 278 000 tweets and achieved 89.80% accuracy. The highest 91.47% accuracy is achieved by the proposed model which is higher than the existing techniques [1, 11, 12, 14]. The word cloud about climate change is shown in Fig. 3. It is a visual representation of text which shows most common words used in a text about climate change. The size of each word indicates how frequently it appears in the given text. The largest words are "climate change," "global warming," "CO₂," "emissions," "temperature," "weather," "impact," "sea level," "crisis," and "adaptation". These words suggest that the text is about the causes, impacts, and solutions of climate change. Word clouds are useful for understanding the terms and phrases of climate change and gaining insight into public perceptions.

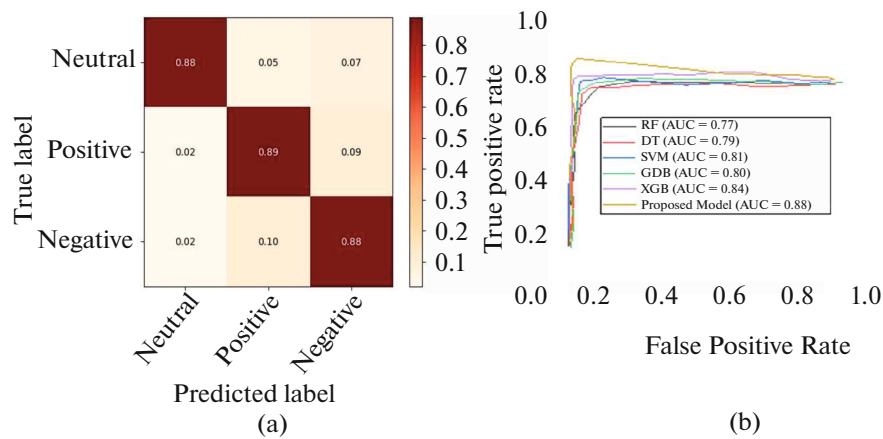


Fig. 6. (a) Confusion matrix and (b) ROC curve of classifiers on Climate Change Sentiment data set.

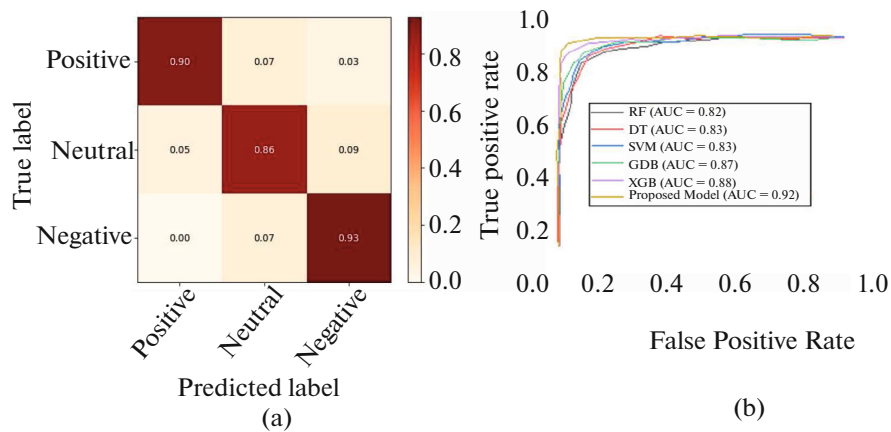


Fig. 7. (a) Confusion matrix and (b) ROC curve of classifiers on Twitter tweets data set.

model correctly classified 88, 89, and 88% of the positive, neutral, and negative samples, respectively. The proposed model correctly classified 90, 86, and 93% of the positive, neutral, and negative samples, respectively on Twitter dataset.

The ROC curves for different machine learning models on four different datasets are depicted in Figs. 4b, 5b, 6b, and 17b, respectively. A ROC curve that is closer to the top-left corner indicates a better model performance. The AUC (area under the curve) measures the model's overall performance, and a higher AUC indicates better performance.

According to the results, the proposed model has achieved the best performance of the five models across all four datasets. The proposed model has the highest AUC of 0.86, 0.87, 0.88, 0.92 on Weather Sentiment, Earth Hour 2015, Climate Change Sentiment, and Twitter dataset, respectively. This suggests that the proposed model is the most effective machine learning model for sentiment analysis on all four used datasets in this work.

5. CONCLUSIONS

This paper proposes a stake ensemble machine learning model for analyzing climate change-related tweet sentiments. The proposed model has been evaluated on the climate change-related tweets dataset. It is compared to the other state-of-the-art models, achieving the highest precision, recall, F1-score, and accuracy of 89.72, 91.65, 90.23, and 91.47%, respectively, among all models. The findings of this study show that the proposed model is an effective way to classify climate change-related tweets. Our model can be used to develop effective tools for monitoring and analyzing public opinion on climate change. Performance of the proposed model will be further improved by using different ensemble methods in the future.

Additionally, these tools can be used to identify trends in public opinion about climate change, such as changes in support for climate change policies. Furthermore, based on our findings, these tools can be used to identify specific groups of people who are more or less likely to support climate change policies. With the help of this information, more targeted and effective communication campaigns can be developed.

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DATA AVAILABILITY

This study used the Python library to collect tweets from Twitter.

CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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